Standing wave solution for the generalized Jackiw-Pi model

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Contents

The Lagrangian

- The Lagrangian of the generalized Jackiw-Pi model (L. Sourrouille, 2012, Physical Review D)

$$\mathcal{L} = \frac{1}{2}\epsilon^{\mu\nu\rho}A_\mu\partial_\nu A_\rho + ig(|\phi|^2)\bar{\phi}D_0\phi - \frac{1}{2}|D_i\phi|^2 - V(|\phi|^2)$$

- $g(|\phi|^2) = |\phi|^{2m}, \quad V(|\phi|^2) = \frac{\lambda}{2}|\phi|^{2m+4}$

  matter field: $\phi : \mathbb{R}^{1+2} \to \mathbb{C}$

  gauge field: $A_\mu : \mathbb{R}^{1+2} \to \mathbb{R}$

  covariant derivative: $D_\mu = \partial_\mu + iA_\mu \quad , \mu = 0, 1, 2$

  strength of interaction potential $\lambda > 0$, m is a non-negative integer

EOM

- The equation of motion

$$i(m+1)|\phi|^{2m}D_0\phi + (D_1D_1 + D_2D_2)\phi + \frac{\lambda}{2}(m+2)|\phi|^{2m+2}\phi = 0, \tag{1}$$

$$\partial_0 A_1 - \partial_1 A_0 = -\mathsf{Im}\big(\bar{\phi}D_2\phi\big), \tag{2}$$

$$\partial_0 A_2 - \partial_2 A_0 = \mathsf{Im}\big(\bar{\phi}D_1\phi\big), \tag{3}$$

$$\partial_1 A_2 - \partial_2 A_1 = -\frac{1}{2}|\phi|^{2m+2}. \tag{4}$$

Relate results about JP model

- The system (1)–(4) with $m = 0$: Jackiw-Pi model
  R. Jackiw and S. Pi 1990 Phys. Rev. D, Phys. Rev. Lett.
- Standing wave solution:
  Byeon, Huh and Seok 2012 JFA
- Nontrivial solitary wave solution:
  Jin and Seok 2021 JDE
- Self-dual equations:
  Huh 2020 Rep. Math. Phys.

Properties

- Gauge transformation:

$$\phi \to \phi e^{i\chi}, \qquad A_\mu \to A_\mu - \partial_\mu \chi,$$

$\chi : \mathbb{R}^{1+2} \to \mathbb{R}$.

- Solution of the system:
  $(\phi e^{i\chi},\ A_0 - \partial_0 \chi,\ A_1 - \partial_1 \chi,\ A_2 - \partial_2 \chi)$

- Coulomb gauge condition

$$\partial_1 A_1 + \partial_2 A_2 = 0$$

Standing wave solution

- Standing wave solution to (1)–(4) of the form

$$\begin{aligned} &\phi(t,x) = u\big(|x|\big)e^{i\omega t}, \qquad A_0(t,x) = k\big(|x|\big), \\ &A_1(t,x) = \frac{x_2}{|x|^2}h\big(|x|\big), \qquad A_2(t,x) = -\frac{x_1}{|x|^2}h\big(|x|\big), \end{aligned} \tag{5}$$

- $\omega > 0$ and u, h, k are real valued functions and $h(0) = 0$
- Nonlocal semi-linear elliptic equation:

$$\begin{aligned} &\Delta u - (m+1)(\omega+\xi)u^{2m+1} - (m+1)\left(\int_{|x|}^{\infty}\frac{h(s)}{s}u^2(s)ds\right)u^{2m+1} \\ &\qquad - \frac{h^2(|x|)}{|x|^2}u + \frac{\lambda}{2}(m+2)|u|^{2m+2}u = 0, \end{aligned} \tag{6}$$

where $\xi \in \mathbb{R}$ is a constant and

$$h(s) = \int_0^s \frac{l}{2}u^{2m+2}(l)dl, \qquad k(|x|) = \xi + \int_{|x|}^{\infty}\frac{h(s)}{s}u^2(s)ds$$

Euler-Lagrange equation

- Show that (6) is Euler-Lagrange equation of the functional

$$J(u) \equiv \frac{1}{2}\int_{\mathbb{R}^2} \left|\nabla u\right|^2 + (\omega+\xi)u^{2m+2} + \frac{u^2}{|x|^2}h^2\,dx - \frac{\lambda}{4}\int_{\mathbb{R}^2}\left|u\right|^{2m+4}dx, \quad (7)$$

where $u \in H_r^1$ is a radially symmetric function in $H^1(\mathbb{R}^2)$

- Show that $J \in C^1(H_r^1)$ and a critical point u of J produces a standing wave $(\phi,\, A_0,\, A_1,\, A_2)$ of the form (5)

Theorem (Jin and Huh 2023)

We have three cases depending on the interaction strength λ*.*
(1) *For* $\lambda \in (0,1)$ *and* $\omega > 0$*, there exists no nontrivial standing wave solution* $(\phi,\, A_0,\, A_1,\, A_2)$ *of the form* (5) *satisfying equations* (1)–(4)*.*
(2) *For* $\lambda = 1$ *and* $\omega > 0$*, any standing wave solution* $(\phi,\, A_0,\, A_1,\, A_2)$ *of the form* (5) *satisfying equations* (1)–(4) *and* $|\phi| > 0$ *in* $\mathbb{R}^2$ *has following form*

$$\phi = \left(\frac{\sqrt{8}l}{1+(m+1)|lx|^2}\right)^{\frac{1}{m+1}} e^{i\omega t}, \qquad A_0 = \frac{1}{2}\left(\frac{\sqrt{8}l}{1+(m+1)|lx|^2}\right)^{\frac{2}{m+1}} - \omega,$$

$$A_1 = \frac{2l^2x_2}{1+(m+1)|lx|^2}, \qquad A_2 = -\frac{2l^2x_1}{1+(m+1)|lx|^2}.$$

(3) *For* $\lambda > 1$ *and* $\omega > 0$*, there exists a standing wave solution* $(\phi,\, A_0,\, A_1,\, A_2)$ *of form* (5) *satisfying equations* (1)–(4) *and* $|\phi| > 0$ *in* $\mathbb{R}^2$*. Moreover, if their frequencies* ω *are different, then any two standing waves of the form* (5) *are not gauge equivalent each other.*

Two inequalities

Lemma 1:

We consider the following Gagliardo-Nirenberg inequality

$$\|u\|_{L^{2m+2}(\mathbb{R}^2)} \leq C\|\nabla u\|_{L^2(\mathbb{R}^2)}^{\frac{m}{m+1}}\|u\|_{L^2(\mathbb{R}^2)}^{\frac{1}{m+1}}.$$

Lemma 2:

For any $u \in H_r^1$, the following well-known inequality called Strauss inequality holds

$$|x|^{\frac{1}{2}}|u(|x|)| \leq C\|u\|, \quad |x| > 0,$$

where $\|u\|^2 = \int_{\mathbb{R}^2} |\nabla u|^2 + |u^2| dx$ denotes the Sobolev norm on $H_r^1(\mathbb{R}^2)$.

Some Propositions

Proposition 1

If u is in $H_r^1(\mathbb{R}^2) \cap L_{loc}^\infty(\mathbb{R}^2)$, then A_0 is in $L^\infty(\mathbb{R}^2)$. Furthermore, if u is in $H_r^1(\mathbb{R}^2) \cap C(\mathbb{R}^2)$, then A_0, A_1 and A_2 are in $L^\infty(\mathbb{R}^2) \cap C^1(\mathbb{R}^2)$.
Then, we can apply a variational argument to obtain a solution of Eq. (6). In fact, (6) is the Euler-Lagrange equation of the following functional

$$J(u) \equiv \frac{1}{2} \int_{\mathbb{R}^2} \left|\nabla u\right|^2 + (\omega + \xi) u^{2m+2} + \frac{u^2}{|x|^2} h^2(|x|) dx - \frac{\lambda}{4} \int_{\mathbb{R}^2} \left|u\right|^{2m+4} dx,$$

for $u \in H_r^1(\mathbb{R}^2)$.

Proposition 2

The functional J is continuously differentiable on H_r^1 and its critical point u is a weak solution of (6). Furthermore, a critical point u of J belongs to $C^2(\mathbb{R}^2)$, so the weak solution u is a classical solution of (6).

Pohozaev identity

Proposition 3

Let b, c and d be real constants and $u \in H_1^r(\mathbb{R}^2)$ be a weak solution of the equation:

$$\Delta u + b(m+1)u^{2m+1} + c(m+1)\left(\int_{|x|}^{\infty} \frac{h(s)}{s} u^2(s)ds\right) u^{2m+1}$$
$$+ c\frac{h^2(|x|)}{|x|^2} u + d(m+2)|u|^{2m+2}u = 0 \quad \text{in } \mathbb{R}^2,$$

where $h(s) = \int_0^s \frac{r}{2} u^{2m+2} dr$. Then, there holds the following integral identity

$$b\int_{\mathbb{R}^2} u^{2m+2}dx + 2c\int_{\mathbb{R}^2} \frac{h^2(|x|)}{|x|^2} u^2 dx + d\int_{\mathbb{R}^2} |u|^{2m+4} dx = 0.$$

Proposition 4

For $u \in H_r^1(\mathbb{R}^2)$, the following inequality holds:

$$\int_{\mathbb{R}^2} \left|u(|x|)\right|^{2m+4} dx$$

$$\leq 4 \left(\int_{\mathbb{R}^2} \left|\nabla u\right|^2 dx\right)^{1/2} \left(\int_{\mathbb{R}^2} \frac{\left|u(|x|)\right|^2}{|x|^2} \left(\int_0^{|x|} \frac{s}{2} u^{2m+2}(s) ds\right)^2 dx\right)^{1/2}.$$

Furthermore, the equality is attained by a continuum of functions

$$\left\{ u_l = \frac{(\sqrt{8}l)^{\frac{1}{m+1}}}{(1+(m+1)l^2|x|^2)^{\frac{1}{m+1}}} \,\middle|\, l \in (0, +\infty) \right\}$$

The case of $\lambda \in (0,1)$

- Suppose that for given $\lambda \in (0,1)$ and $\omega > 0$, $\phi = (ue^{i\omega t},\, A_0,\, A_1,\, A_2)$ is a nontrivial standing wave solution
- u is a solution of the equation

$$\begin{aligned}\Delta u - (m+1)(\omega+\xi)u^{2m+1} - (m+1)\int_{|x|}^{\infty} \frac{h(s)}{s}u^2(s)dsu^{2m+1} \\ - \frac{h^2(|x|)}{|x|^2}u + \frac{\lambda}{2}(m+2)|u|^{2m+2}u = 0\end{aligned} \tag{8}$$

- By the Pohozaev identity, we have

$$(\omega+\xi)\int_{\mathbb{R}^2} u^{2m+2}dx + 2\int_{\mathbb{R}^2} \frac{h^2(|x|)}{|x|^2}dx - \frac{\lambda}{2}\int_{\mathbb{R}^2} |u|^{2m+4}dx = 0$$

- We multiply (8) by u

$$\int_{\mathbb{R}^2} |\nabla u|^2 + (m+1)(\omega+\xi)u^{2m+2} + (2m+3)\frac{h^2(|x|)}{|x|^2}u^2 - \frac{\lambda}{2}(m+2)|u|^{2m+4}dx = 0$$

The case of $\lambda \in (0,1)$

- Combining the above two identities

$$\hat{J}(u) = \frac{1}{2}\left(\int_{\mathbb{R}^2} |\nabla u|^2 + \frac{u^2}{|x|^2}h^2(|x|) - \frac{\lambda}{2}|u|^{2m+4}dx\right) = 0$$

- If $\lambda < 1$, we have

$$\begin{aligned}
\hat{J}(u) =& \frac{1}{2}\left(\int_{\mathbb{R}^2} |\nabla u|^2 + \frac{u^2}{|x|^2}h^2(|x|) - \frac{\lambda}{2}|u|^{2m+4}dx\right) \\
>& \frac{1}{2}\int_{\mathbb{R}^2} |\nabla u|^2 + \frac{u^2}{|x|^2}h^2(|x|)dx - \left(\int_{\mathbb{R}^2} |\nabla u|^2 dx\right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^2} \frac{u^2}{|x|^2}h^2(|x|)dx\right)^{\frac{1}{2}} \\
=& \frac{1}{2}\left[\left(\int_{\mathbb{R}^2} |\nabla u|^2 dx\right)^{\frac{1}{2}} - \left(\int_{\mathbb{R}^2} \frac{u^2}{|x|^2}h^2(|x|)dx\right)^{\frac{1}{2}}\right]^2 \geq 0
\end{aligned}$$

  for every $u \neq 0$

- Contradicts with the existence of a nontrivial solution u of (8) when $\lambda < 1$

The case of $\lambda = 1$

- By the Proposition 4, we know that for each $l > 0$, the function u_l is a minimizer of $\hat{J}$. Note that the Euler-Lagrange equation of $\hat{J}$ is

$$\Delta u - (m+1)\int_{|x|}^{\infty} \frac{h(s)}{s} u^2(s) ds u^{2m+1} - \frac{h^2(|x|)}{|x|^2} u + \frac{\lambda}{2}(m+2)|u|^{2m+2} u = 0,$$

- Next, we show that for $\lambda = 1$, any standing wave solution $(\phi,\, A_0,\, A_1,\, A_2)$ of the form (5) with $|\phi| > 0$ is exactly given by the following explicit formula:

$$(\phi,\, A_0,\, A_1,\, A_2) = \left(\left(\frac{\sqrt{8}l}{1+(m+1)|lx|^2} \right)^{\frac{1}{m+1}} e^{i\omega t}, \right.$$
$$\left. \frac{1}{2}\left(\frac{\sqrt{8}l}{1+(m+1)|lx|^2} \right)^{\frac{2}{m+1}} - \omega, \frac{2l^2 x_2}{1+(m+1)|lx|^2}, -\frac{2l^2 x_1}{1+(m+1)|lx|^2} \right)$$

The case of $\lambda = 1$

- Suppose that (ϕ, A_0, A_1, A_2) is a standing wave solution of the form (5) and $|\phi| > 0$.
- We see from elementary calculations that

$$\begin{aligned}
&\frac{1}{2}\int_{\mathbb{R}^2} |\partial_1 u|^2 + |\partial_2 u|^2 + (A_1^2 + A_2^2)u^2 dx - \frac{1}{4}\int_{\mathbb{R}^2} |u|^{2m+4} dx \\
=&\frac{1}{2}\int_{\mathbb{R}^2} (\partial_1 u - A_2 u)^2 + (\partial_2 u + A_1 u)^2 dx + \frac{1}{2}\int_{\mathbb{R}^2} \partial_1(A_2 u^2) + \partial_2(-A_1 u^2) dx \\
&+ \frac{1}{2}\int_{\mathbb{R}^2} (\partial_2 A_1 - \partial_1 A_2)u^2 dx - \frac{1}{4}\int_{\mathbb{R}^2} |u|^{2m+4} dx.
\end{aligned}$$

- We recall that $\partial_1 A_2 - \partial_2 A_1 = -\frac{1}{2}|u|^{2m+2}$, then we have

$$\hat{J}(u) = \frac{1}{2}\int_{\mathbb{R}^2} (\partial_1 u - A_2 u)^2 + (\partial_2 u + A_1 u)^2 dx.$$

The case of $\lambda = 1$

- By $\hat{J} = 0$, we have the following relations

$$\partial_1 u = A_2 u, \qquad \partial_2 u = -A_1 u.$$

Then by a change of variable $v = 2(m+1)\log u$, the above relation is equivalent to

$$\partial_1 v = 2(m+1)A_2, \qquad \partial_2 v = -2(m+1)A_1$$

which imply that

$$\Delta v + (m+1)e^v = 0.$$

- Note that v solves the following initial value problem

$$v'' + \frac{1}{r}v' + (m+1)e^v = 0, \qquad v(0) = 2(m+1)\log u(0), \qquad v'(0) = 0.$$

The case of $\lambda = 1$

- We can check that the I.V.P is solved by the following solution

$$v(r) = -2\log(\frac{(m+1)u^{2m+2}(0)}{8} + 1) + 2(m+1)\log u(0)$$

  and recall that I.V.P has a unique solution.

- We arrive at

$$u \equiv \left(\frac{\sqrt{8}l}{1 + (m+1)|lx|^2} \right)^{\frac{1}{m+1}},$$

  where $l^2 = \frac{u^{2m+2}(0)}{8}$.

- Then, we can directly calculate A_0, A_1 and A_2 from u to obtain the explicit formula of (ϕ, A_0, A_1, A_2).

Sketch of proof ($\lambda > 1$)

- We introduce the following constrained minimization problems

$$I = \inf_{u \in M} \int_{\mathbb{R}^2} u^{2m+2} dx, \qquad M = \{u \in H_r^1(\mathbb{R}^2) \backslash \{0\} | \hat{J}(u) = 0\}. \tag{9}$$

- Prove the constraint M is nonempty
- The minimization problem (9) lacks some compactness (Renormalization argument)
- We show that there exitst nonegative function $u_0 \in H_r^1(\mathbb{R}^2)$, such that u_0 is the minimizer.

Thank you for your attention and Happy Birthday to Professor Yu!